Extractive-Abstractive Unified Model for Related Work Generation

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Abstract

Related work can help readers quickly understand the context of a study. In recent years, many researchers have been working on automating this process using abstractive approaches. With the development of the Transformer model and the attention mechanism, it is rare to see extractive models in related work generation tasks. However, since the computation time of the attention mechanism is greatly affected by the length of the input sequence, the input length of the abstractive model is often limited. Also, the current related work generation model training is time-consuming, leading to slower model iterations. In this paper, we propose the Extractive-Abstractive Unified Related Work Generation Model (EAU) to support long context input. The cross-attention based extractive model optimization algorithm can optimize the extractive model in the generation task. To achieve fast model iteration and training, we construct a large language model-assisted enhanced related work generation dataset, Delve-aug, which is less than one-tenth the size of the original dataset and can reduce the training time by more than ten times. A large number of experiments show that Delve-aug can improve performance by reducing training time. Moreover, the EAU model performs better than the state-of-the-art baseline models and can generate higher-quality related works.

1 Introduction

High-quality related work usually requires citing and interpreting a large number of references, which requires authors to spend a great deal of time reading and summarizing, as well as keeping up to date with the latest research in the field (Hu and Wan, 2014). In recent years, the emergence of Related work generation tasks has attracted more and more attention from researchers concerned with text summarization (Chen et al., 2021, 2022).

In recent years, researchers have carried out many attempts through abstractive text generation methods. For example, Xing et al. (2020a) firstly combined the context of citations with the abstracts of the cited papers as the input to the generation model of Related work. Ge et al. (2021) encoded the citation network through Graph Attention Networks (GAT) (Veličković et al., 2017) and utilized it as the external knowledge in the generation process. Chen et al. (2022) proposed a target-aware Related work generation model that captures the relations between references and target papers through a target-centered attention mechanism. Through well-designed encoding strategies (Chen et al., 2021), external knowledge (Ge et al., 2021), or novel training techniques (Chen et al., 2022), these studies have made good progress in generating related works with smooth transitions and informative content.

With the development of Transformer models and attention mechanisms in recent years(Vaswani et al., 2017), extractive models are rarely seen in text summarization tasks. However, compared to abstractive models, extractive models still have several advantages in terms of the length of the context window, the accuracy of outputting specific content, and the speed of training and inference since they are essentially performing the text classification process, which is not the case with abstractive models.

In the related work generation task, although the adopted dataset contains the abstract content of the cited papers, restricted by the input length limitation of the abstractive model, the common practice is to splice the fixed-length content from the header of each document as the input. However, this oversimplified approach will undoubtedly result in a large amount of information being lost in the data samples. Therefore, the proposed model in this paper will attempt to merge the extractive and abstractive models into one model, where the output of the extractive model will be used as the input to

the abstractive model, and the final result will be produced by the abstractive model. This approach utilizes the extractive model to fully mine the information in the input samples and feeds the valuable information into the abstractive model. The abstractive model can further explore and output varied and coherent related work content.

Meanwhile, the training duration in the related work generation task is longer than that for tasks such as text classification due to the longer input and larger dataset size. In order to shorten the training time and speed up the model iteration, we will also try to utilize the capability of the large language models to construct an enhanced small batch related work generation dataset, which is less than one-tenth of the original size. It will significantly accelerate the model training process and can improve the model effect as well.

The contributions of this paper can be categorized into the following three aspects:

(1) Proposes Extractive-Abstractive Unified Related Work Generation Model (EAU), which fuses the extractive and abstractive models to support long input context without compromising computation time.

(2) Proposes the cross-attention based extractive model optimization algorithm that uses the differences between the distributions of classification probability and the cross-attention scores to compute the classification error. Eventually achieve the optimization of the extractive model.

(3) The Delve-aug dataset is constructed with the aid of a large language model, which contains a total of 5000 samples and is 6% of the original size. Experiments can indicate that the Delve-aug dataset can speed up the training process and bring about an improvement in the model results at a smaller scale.

2 Related Works

Abu Ra'ed et al (AbuRa'ed et al., 2020) used a simple sequence-to-sequence approach and applied implementation to the task of generating citation sentences given the title and abstract of a cited paper. Xing et al. (2020b) takes the citation context along with the abstracts of the cited papers as input to generate the citation text. The essence of this research is that they expect to generate the corresponding citation content by learning the context of the citation content. However, their method can only generate citations for a single article. Chen et al. (2021) took the abstracts of multiple cited papers as input and constructed a relation graph between these papers during the encoding process, and then iteratively updated it during the encoding process through a specific module, which was ultimately used as the auxiliary information for the decoding process of generation. Chen et al. (2022) used the paper to which the final obtained related work belongs as the target, and used a target-centered attention mechanism to generate related work with higher quality content. in addition to information from references, they used keywords that could represent the target paper as input. Liu et al. (2023) use causal intervention to prompt the model to get rid of spurious correlations in the generation process and learn the causal relations. The final generated content has better coherence.

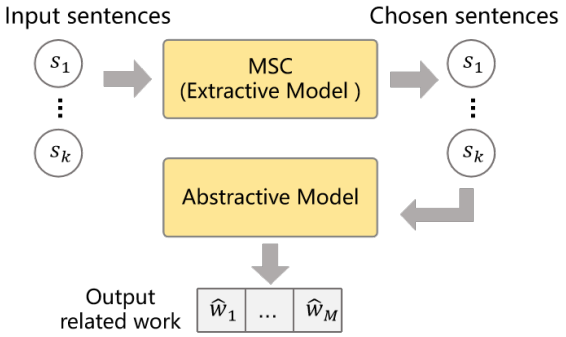


Figure 1: The overall architecture of EAU.

3 Methodology

3.1 Problem Definition

Given a set of references $D = \{r_1, ..., r_{|D|}\}$, denote the true Related work as $Y = (w_1, w_2, ..., w_M)$, where $r_i = (s_1^i, s_2^i, ..., s_{|r_i|}^i)$ denotes a single cited paper, s_j^i is the jth sentence in r_i, and w_j is the jth word in related work Y. Unlike the previous practice of cutting and splicing inputs that are out of length, the EAU model proposed in this paper receives the complete D as input. Specifically, the extractive model in the EAU will first output a subset $\tilde{D}$ of D and give it as input to the abstractive model, which will then output the final related work result $\hat{Y} = (\hat{w}_1, \hat{w}_2, ..., \hat{w}_{\hat{M}})$. The model is optimized by minimizing the difference between Y and $\hat{Y}$.

3.2 Overall Architecture

The Extractive-Abstractive Unified Related Work Generation Model (EAU) differs from the previous models that are only extractive or abstractive but

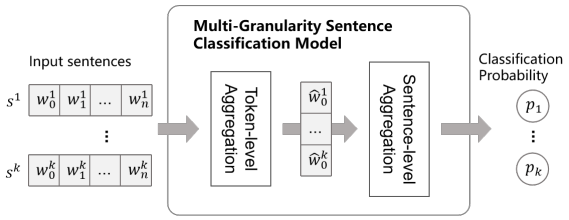


Figure 2: The architecture of MSC model.

is a fusion of these two models, as illustrated in Figure 1. In the model workflow, the extractive model receives the full input of samples first, and in this process, it does not trim the content and extracts the sentences that are considered appropriate by the model. The extracted sentences are concatenated and given to the abstractive model, which then carries out the sequence-to-sequence generation process. The final output of the abstractive model is the output of the whole EAU model.

The EAU model integrates the advantages of the extractive and abstractive models. Firstly, it solves the information loss problem caused by trimming the input content. At the same time, because the extractive model filters the content, the content received by the abstractive model is considered to be of a higher quality and more comprehensive. The abstractive model will further generate a more coherent and varied summarized content. The overall framework can be divided into three parts: the extractive model, the abstractive model and the cross-attention based extractive model optimization algorithm.

3.3 Multi-Granularity Sentence Classification Model

Multi-granularity Sentence Classification (MSC) model is involved in training and iterative optimization as the extractive model in the overall model. The general structure of the MSC model is shown in Figure 2. In MSC, the aggregation of text information is divided into two granularities, i.e., token-level aggregation and sentence-level aggregation. Assume that the total number of sentences present in a single document is k, and denote each sentence as a sequence $s^i = (x_0^i, x_1^i, x_2^i, ..., x_n^i)$, where i is the sentence sequence number and n denotes the total length of the sentence, then the process of token-level aggregation can be represented as

$$\hat{s} = \mathrm{LangModel}_0(s) \quad (1)$$

$$\mathrm{LangModel}(x) = \mathrm{LN}(x + \mathrm{MHA}(x, x, x)) \quad (2)$$

$$\mathrm{MHA}(x, x, x) = \mathrm{Concat}(\mathrm{head}_1, ..., \mathrm{head}_\mathrm{i}) \quad (3)$$

$$\mathrm{head}_\mathrm{i}(x, x, x) = \mathrm{softmax}(\frac{x^T \cdot x}{\sqrt{d_k}}) \cdot x \quad (4)$$

where $\mathrm{LangModel}(\cdot)$ can be regarded as a regular pre-trained language model, $\mathrm{LN}(\cdot)$ is the layer normalization process, and $\mathrm{MHA}(\cdot)$ is the multi-head attention process. The final sentence after token-level aggregation update can be represented as $\hat{s}^i = (\hat{x}_0^i, \hat{x}_1^i, \hat{x}_2^i, ..., \hat{x}_n^i)$.

During sentence-level aggregation, the updated initial word representations of the k sentences are concatenated to obtain a sequence of length k, denoted as $s' = (\hat{x}_0^1, \hat{x}_0^2, \hat{x}_0^3, ..., \hat{x}_0^k)$. Similarly, the computational process of sentence-level aggregation can be represented as

$$p = \mathrm{Linear}(\mathrm{LangModel}_1(s')) \in R^{k \times 2} \quad (5)$$

The model output p is a binary categorical probability distribution of length k and width 2. A set of distributions of length 2 at each position indicates the probability that the corresponding sentence is a positive and negative example.The final computational procedure of the MSC model can be represented as follows

$$\hat{s} = \mathrm{LangModel}_0(s) \quad (6)$$

$$s' = \hat{s}[:, 0] \quad (7)$$

$$p = \mathrm{Linear}(\mathrm{LangModel}_1(s')) \in R^{k \times 2} \quad (8)$$

where $s' = \hat{s}[:, 0]$ denotes the indexing process that takes out the first element of all its rows against $\hat{s}$.

3.4 Cross-attention based Extractive Model Optimization Algorithm

For the EAU model to be trained stably and efficiently, the key problem that needs to be solved is the optimization process of the extractive model. Neither the generated content, data samples, nor the corresponding losses of the abstractive model can be directly used to optimize the extractive model. The underlying reason is that the process of splicing the classification results of the extraction model to get the inputs of the abstractive model leads to the cross-entropy loss of the abstractive model not being able to be back-propagated to the extractive model.

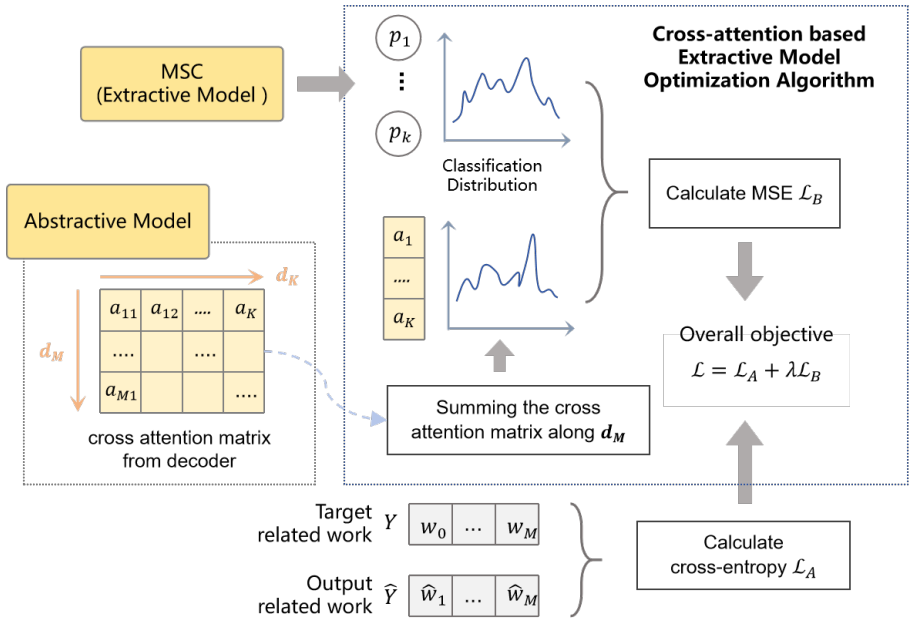


Figure 3: The Cross-attention based Extractive Model Optimization Algorithm

In order to solve the optimization problem of the extractive model, we focus our attention on the matrix of cross-attention scores in the decoder of the abstractive model. As shown in Figure 1, two main distributions are used in the optimization algorithm. One is the classification output p of the extractive model, as shown in Equation 5, and the other one is the cross-attention scores in the decoder.We believe that sentences with a higher probability of positive examples in the classification process should accordingly receive more attention in the generation process, and thus, the classification probability distribution output by a good extractive model should be similar to the attention received by the input content in the abstractive model.

In order to characterize the degree of attention that the input content receives during the decoding process, it is necessary to first recall the meaning of cross attention scores. If there is a matrix of cross attention scores $M_{crs} \in \mathbb{R}^{M\times K}$, then M stands for the length of the decoded content and K for the length of the input content. For the element $a_{ij} = M_{crs}[i, j]$, the meaning is the attention score of the i-th word in the decoded sequence with respect to the j-th word in the input sequence. If the first dimension of M_{crs} is denoted as d_M and the second dimension is denoted as d_K, then the result of retaining the d_K dimension for summing all scores along the direction of d_M, p_{crs}, is the attention scores that the input content of length K is subjected to during decoding, respectively. Thus, the optimization process of the extractive model can be achieved by reducing the difference between p_{crs} and p. This process can be formulated as

$$p_{crs} = \text{sum}_{axis=0}(M_{crs}) \in \mathbb{R}^{1\times K} \quad (9)$$

$$\mathcal{L}_B = \text{MSE}(p_{crs}, \text{Linear}(p)) \quad (10)$$

where $\text{MSE}(\cdot)$ is the mean square error, which is used to compute the difference between the two distributions, and $\text{Linear}(\cdot)$ is the linear projection process, which is used to achieve consistency between the lengths of p and p_{crs}. The $\mathcal{L}_B$ is the final objective function of the extractive model.

Denote the cross-entropy objective function of the abstractive model as

$$\mathcal{L}_A = -\sum_{i}^{\hat{M}} P_i(\mathbf{Y}) \cdot \log P_i(\hat{\mathbf{Y}}) \quad (11)$$

Then the objective function of the EAU model is denoted as

$$\mathcal{L} = \mathcal{L}_A + \lambda\mathcal{L}_B \quad (12)$$

where λ is a hyperparameter that controls the proportion of the extractive model loss.

3.5 Large Language Model-Assisted Enhanced Dataset

In this paper, a total of three large language model-assisted related work generation datasets will be obtained where the Delve-aug dataset can be directly

used for model training and serve to accelerate training and improve performance, the Delve-cls dataset is used to assist in pre-training the MSC model that presented in Section 3.3, and the Delve-prob dataset has the original the full scale of the Delve dataset, and is labeled with binary classification probabilities for each input sentence in the data samples by invoking the trained MSC model.

In the process of making the Delve-aug dataset, the original Delve dataset will first be sampled to obtain a subset with a size of 5000. The input will be spliced with the prompt *"Select four sentences from this paragraph that represent contributions"* and fed into ChatGLM3(Du et al., 2022), and following the given prompt, ChatGLM3 will output four sentences that can summarize the contribution. By invoking the large language model for content extraction, the length of the input can be greatly reduced, and it can be found that the extracted content is not strictly consistent - some text errors introduced during the data preprocessing stage are corrected, which in turn further enhances the quality of the data.

To evaluate the effect of Delve-aug, there still exist two problems: **(1)** previous models were evaluated on the full test set of the Delve, whereas Delve-aug has only 5000 samples, so there is a lack of consistent test set with the Delve-aug. **(2)** Since Delve does not contain any knowledge of the large language model, it is highly likely that the model trained on Delve-aug will not achieve the best results on Delve's test set. The most intuitive solution to these two problems is to use the large language model to augment the test set of Delve. However, considering the time cost of model reasoning, we choose to construct the Delve-cls dataset based on the Delve-aug dataset, and annotate the Delve test set with the MSC model that pre-trained on Delve-cls, and finally obtain the dataset Delve-prob that can evaluate the training effect of Delve-aug.

4 Experiments

4.1 Datasets and Settings

Datasets The datasets used are Delve (Akujuobi and Zhang, 2017), Delve-aug and Delve-prob. The Delve dataset is a publicly available related work generation dataset proposed by Chen et al.(Chen et al., 2021), Delve-aug and Delve-prob are both datasets based on large language model enhancement obtained in Section 3.5. Delve-aug is mainly used for model training, which can lead to faster training speed. The test set of Delve-prob is kept the same scale as the Delve test set and has an additional input sentence classification probability. Thus, by using Delve-aug as a training set and Delve-prob's test set for testing, it is possible to evaluate Delve-aug's ability to speed up and enhance performance on related work generation tasks under fair conditions. The statistics of these three datasets are shown in Table 1.

Settings The experiments are implemented based on PyTorch and NVIDIA 3080Ti GPU. In the experiments verifying the Delve-aug dataset, the maximum number of reference papers for each sample is set to 5, and the first $440/|D|$ words of the abstracts of each reference paper are selected to be spliced to get the model input. In the experiments to validate the EAU model, the inputs are sentence sequences and are not truncated. The total number of sentences in the generated related work is set to 6, i.e., $s = 6$. The maximum length of the related work $\hat{y}$ is fixed to 220. The word embedding dimension of the abstractive model is 768, and both the number of attention heads and the number of layers are set to 12. An Adam algorithm is used as the optimizer, and the learning rate is set to 0.01.

4.2 Baselines

We selected 3 extractive generative models and 12 generative models as baseline models.

Extractive (1) **TextRank** (Mihalcea and Tarau, 2004): a graph-based text ranking model for multi-document sentence extraction. (2) **BertSumEXT** (Liu and Lapata, 2019): an extractive summarization model that extends BERT by inserting multiple [CLS] tokens and uses their representations to classify sentences. (3) **MGSum-ext** (Jin et al., 2020): a multi-granularity network that jointly learns different semantic representations of words, sentences and documents.

Abstractive (1)**TransformerABS**(Vaswani et al., 2017): a generative summarization model based on Transformer. (2)**BertSumABS**(Liu and Lapata, 2019): BERT-based generative model with two-stage fine-tuning approach. (3)**MGSum-abs**(Jin et al., 2020): a multi-granularity interaction network for generative document summarization. (4)**GS**(Li et al., 2020): a generative summarization model that utilizes a specific network of graphs to encode documents.(5)**T5-base**(Raffel et al., 2020): a commonly used text-to-text generative model that utilizes transfer learning

	Delve	Delve-aug	Delve-prob(Test Set)
#Samples	72,927/3,000/3,000	5,000	3,000
#Cited papers	3.69	3.68	3.69
#Sentence	7.88	11.65	11.64
#Output length	181	232	234
#Vocabulary	190,381	71,729	71,853

Table 1: Statistics of the datasets

techniques for training. (6)**BART-base**(Lewis et al., 2020): a commonly used generative model for text-to-text that combines the advantages of autoregressive training and denoising pre-trained targets. (7)**Longformer**(Beltagy et al., 2020): a Transformer-based model that efficiently handles long-distance dependencies in text. (8)**RGG**(Chen et al., 2021): an encoder-decoder model designed for related work generation, which constructs and iteratively optimizes the relation graphs of the references.(9)**NG-Abs**(Zhu et al., 2023): a BART model optimized for cross-entropy loss and proposed differentiable N-gram objective function. (10)**TAG**(Chen et al., 2022): treats the paper to which the related work belongs as a target and uses a target-centered attention mechanism to generate the related work. (11)**BERT-Gen**(Rothe et al., 2020): an encoder-decoder generation model utilizing the BERT model. (12)**CaM**: model introduces causal intervention to learn reliable causal relations and improve generation performance.

Due to the long length of the complete inputs in the dataset, the inference speed of ChatGLM3-7B on a 3080Ti GPU is about $16s/iter$. If it is used as a baseline then the total inference time taken will be more than 10 hours, so we did not use large language models as baselines considering the huge time cost.

4.3 Baseline Results

Table 2 shows the performance of the EAU model and baseline models on the Delve and Delve-aug datasets. It can be found that the model trained with Delve-aug outperforms the model trained with Delve. In particular, analyzing experiments 14 vs. 17 and 15 vs. 16 reveals that with identical models, the fact that Delve-aug has only 5000 samples makes the training time for a single epoch reduce between 10 and 20 times, and on top of that brings further performance improvements in ROUGE scores.

Analyzing the experiments based on Delve-aug,it can be found that EAU scores only slightly lower on ROUGE-2 compared to CaM, but achieves a significant advantage on both ROUGE-1 and ROUGE-L. EAU also has a faster training speed compared to CaM. Compared with the results based on the Delve, the advantages of EAU in ROUGE-1 and ROUGE-L are further extended, and only ROUGE-2 slightly lags behind TAG and CaM. The above experiment results show that EAU can effectively solve the information loss problem caused by truncation of input content. Moreover, the cross-attention based extractive model optimization algorithm can effectively optimize the extractive model during the training process.

4.4 Ablation Study

The ablation experiment mainly focuses on verifying the effectiveness of the extraction model optimization algorithm in EAU. When it is disabled, the output of the extractive model will no longer change with the training process, i.e., the inputs received by the abstractive model have been consistent with the initial outputs of the extractive model during each step of training. Thus, the ablation model essentially becomes the abstractive model in EAU. Therefore, the ablation model corresponds to the BERT-Gen model of experiment 17 in Table 2. By analyzing experiments 17 and 18, it can be found that when the optimization of the extractive model is disabled, the effect of the model decreases in all metrics. This comparative result can indicate the validity of the optimization algorithm of the extractive model.

4.5 Parameter Experiment

As shown in Equation 12, the hyperparameter λ value in the final joint optimization loss directly balances the strength of the extractive process and the abstractive process. This section will explore the effect of λ on the performance of EAU.

No.	Model	Delve			
		ROUGE-1	ROUGE-2	ROUGE-L	Training time (secs)
1	TextRank	25.25	3.04	22.14	-
2	BertSumEXT	28.43	3.98	24.71	-
3	MGSum-ext	27.85	3.95	24.28	-
4	TransformerABS	26.89	3.92	23.64	-
5	BertSumABS	28.02	3.50	24.74	-
6	MGSum-abs	28.13	4.12	24.95	-
7	GS	28.27	4.36	25.08	-
8	T5-base	26.38	5.69	24.35	3433
9	BART-base	26.96	5.33	24.42	3201
10	longformer	28.05	5.20	25.65	4059
11	RRG	29.10	4.94	26.29	-
12	NG-Abs	27.49	5.93	24.56	-
13	TAG	27.82	6.16	25.50	-
14	BERT-Gen	29.01	4.94	26.29	3327
15	CaM	29.31	6.17	26.61	4115
	Model	**Delve-aug**			
		ROUGE-1	ROUGE-2	ROUGE-L	
16	CaM	29.44	5.97	27.48	341
17	BERT-Gen	29.60	5.79	27.38	**255**
18	EAU	**29.70**	5.93	**27.56**	283

Table 2: Baseline experiment results

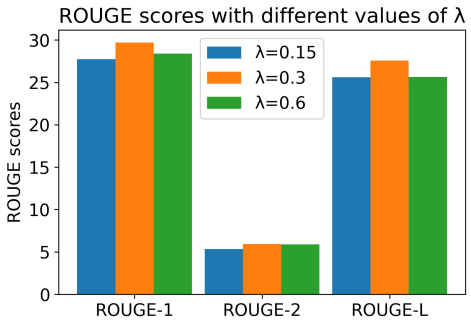
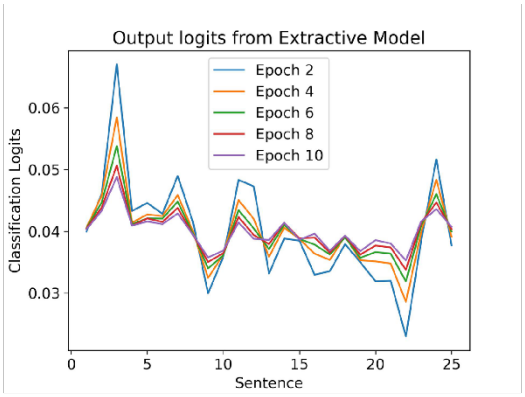
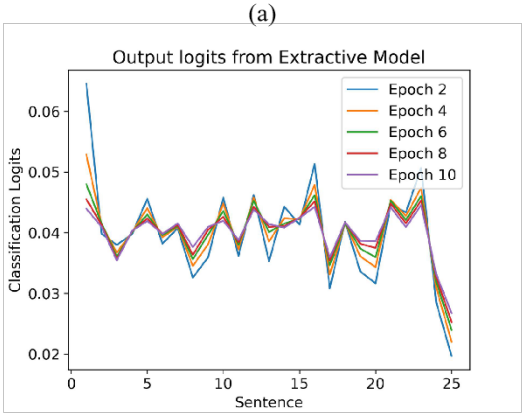


Figure 4: ROUGE scores with different λ values



(a)



(b)

Figure 5: Variation of the extractive model output distribution

Figure 4 shows the ROUGE scores for the EAU model when λ takes values of 0.15,0.3,0.6 respectively. It can be found that the optimal λ value is 0.3. Combined with the analysis of Equation 12, at this point, the ratio of the loss of the abstractive model to the loss of the extractive model is $10:3$. Compared to the abstractive model, the extractive model has a relatively small share of error. This is because the amount of parameters in the abstractive model is much larger than that in the extractive model. If the abstractive model is to converge, the distribution of its inputs should not fluctuate drastically. Therefore, the EAU model is most effective when the error of the extractive model affects the overall model by a relatively small proportion.

case 1	**Discarded**	In this paper, we propose a new algorithm called regularized locality preserving indexing (RLPI).
	Added	It uses fisher 's linear discriminant, a classical tool from statistical pattern recognition, to project training instances to a carefully selected lowdimensional subspace before inducing a decision tree on the projected instances.
	Discarded	Different from latent semantic indexing (LSI) which is optimal in the sense of global euclidean structure, LPI is optimal in the sense of local manifold structure.
	Added	Benefit from recent progresses on spectral graph analysis, we cast the original LPI algorithm into a regression framework which enable us to avoid eigendecomposition of dense matrices.
case 2	**Discarded**	Because of the numerous earthquakes and the large number of twitter users throughout the country, we can detect an earthquake by monitoring tweets with high probability (of earthquakes of japan meteorological agency seismic intensity scale or more are detected).
	Added	Tweets are the most uptodate and inclusive stream of information and commentary on current events, but they are also fragmented and noisy, motivating the need for systems that can extract, aggregate and categorize important events.

Table 3: Sentence discarded and added by the extractive model during training

4.6 Visualization and Case Study

In order to concretely show the changes in the output of the extractive model during the optimization process, two training samples are selected, and the classification distributions from the extraction model in different epochs are plotted. The results are shown in Figure 5, the x-axis is the sentence number, the y-axis is the classification probability.Observing the results, it can be found that the overall variance of the output of the extractive model tends to decrease, i.e., smoother, as the training and optimization proceed. When the classification threshold is set, a small number of certain sentences are replaced in the later process compared to the results in the early stage of training, thus achieving better generation results. This result, on the one hand, shows that the optimization algorithm is able to optimize the extractive model stably and effectively, and on the other hand, it also verifies the reliability of the practice of loading the pre-trained weights when initializing the extractive model as introduced in Section 3.5.

Table 4.5 shows how the sentences from these two samples were discarded and newly adopted by the extractive model. Case 1 shows that the newly adopted sentences focus on elaborating the perspective of the research, while the sentences that simply name the model and those that point out the differences with the other work are discarded. This is consistent with the need to write related works, where the model requires enough details to summarize and analyze related studies, and the relations between a cited work and other studies are not quite important. Case 2 reveals that the discarded content focuses on why the scheme is feasible from the perspective of users. In contrast, the newly adopted content points out the motivation of the scheme from the perspective of the inherent characteristics of social media, with the latter containing significantly more valuable information.

5 Conclusions

In this paper, we propose an Extractive-Abstractive Unified Related Work Generation Model (EAU), which supports longer context input. The core mechanism of the EAU to be trained effectively is the cross-attention based extractive model optimization algorithm, which takes the difference between the sentence classification probability distribution and the cross-attention scores from the abstractive model as the loss of the extractive model.We also constructed a large language model-assisted augmented related work generation dataset Delve-aug.The reduced size greatly shortens the training time and, at the same time, provides a guarantee for the fast iteration of the model. A large number of experiments show that Delve-aug can improve performance by reducing training time by more than 10 times. Moreover, the EAU model performs better than the state-of-the-art baseline models and can generate higher-quality related works.

Limitations

In Section 3.5, the effectiveness and speed of enhancing the dataset is strongly affected by the scale and performance of the chosen large language model, and therefore it is necessary to consider how to improve the efficiency of data generation. Also, the pre-training process for extractive model in the EAU model imposes an additional demand on the overall model training, which is one of the directions for future work.

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